

DHRUV MOTAVAL

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[Portfolio](#) | [GitHub](#) | [LinkedIn](#) | [Hugging Face](#)

TECHNICAL SKILLS

Languages: Python, SQL, JavaScript

Generative AI: LangChain, LangGraph, RAG, OpenAI API, Groq API, Prompt Engineering, Tool Calling

Backend & APIs: FastAPI, REST APIs, API Integration, SQLite, PostgreSQL

AI & Retrieval: ChromaDB, Sentence Transformers, Embeddings, Semantic Search, Vector Databases

ML & NLP: Scikit-learn, XGBoost, Whisper, spaCy, Pandas, NumPy

DevOps & Tools: Docker, Git, Railway, LangSmith, Testing & Debugging

EXPERIENCE

Web Developer Intern - iWebwala

Jan 2026 – Mar 2026

Python, FastAPI, Twilio, Whisper, ElevenLabs, Railway, Netlify, Git

- Championed the full cycle development and launch of a solar installation company's website, including domain registration and Netlify deployment, leading to a 100% increase in online presence.
- Spearheaded development of an AI voice assistant that automated customer interactions for restaurant reservations and takeout orders, reducing average call duration by 3 minutes.
- Launched a conversational AI pipeline on Railway, integrating Twilio Voice, Whisper, and ElevenLabs to automate 50% of initial customer inquiries.

PROJECTS

ExplainOps - AI Incident Analysis Platform *(Hackathon Team Project)*

github.com/dhruv-motaval/ExplainOps

Python, FastAPI, Scikit-learn, Groq, LangChain

- Architected a two-person hackathon solution analyzing IT incidents from simulated ticket, log, and chat streams, leveraging ML scoring and LLM explanations for faster triage.
- Developed an ML urgency scoring pipeline using Scikit-learn and FastAPI, achieving 0.91 F1 and 0.90 ROC-AUC with 1.2 ms average model inference latency.
- Established a feature-branch Git workflow with pull requests, enabling parallel development and reliable code integration across a 2-person team.

RAG Document Q&A System

github.com/dhruv-motaval/rag-document-qa

FastAPI, LangChain, ChromaDB, Sentence Transformers, LangSmith, SQLite

- Developed an end-to-end RAG pipeline integrating PDF parsing, intelligent chunking, embedding generation, and ChromaDB, achieving 25 ms average semantic retrieval latency with a 100% retrieval hit rate across a 30-query benchmark.
- Built FastAPI endpoints for document ingestion and real-time Q&A, adding similarity-threshold validation to reduce hallucinated responses.
- Used LangSmith tracing to monitor retrieval quality and identify and fix a bottleneck in the retrieval step.

Multilingual Call Intelligence Agent

github.com/dhruv-motaval/multilingual-call-classifier

Python, LangGraph, LangChain, Whisper, spaCy, FastAPI, SQLite

- Engineered an agentic workflow using LangGraph StateGraph and tool calling to transcribe multilingual calls, extract entities, and classify urgency with Whisper and structured LLM outputs.
- Implemented tool-calling for ticket creation and follow-up scheduling, with structured LLM outputs served through FastAPI.
- Built an offline validation framework for LangGraph orchestration using mocked Whisper, spaCy, and LLM clients, enabling deterministic end-to-end testing without live API dependencies.

EDUCATION

B.Tech, Artificial Intelligence - Parul Institute of Technology, Vadodara, Gujarat

2023 – 2026

CGPA: 7.12 / 10

Diploma, Computer Engineering - A. Y. DadaBhai Institute of Technology, Kosamba, Gujarat

2019 – 2023

CGPA: 7.91 / 10